

Lecture 17: Intro to the Term Structure

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Intro

- Last lecture: modern asset pricing in high dimensions
- Today: move to the term structure of interest rates
- Goal: build the bond math, the EH benchmark, the classic evidence, and the Diebold-Li forecasting view
- This is an intro lecture, but not a shy one

Why This Lecture?

- Bond markets give us a laboratory for expected returns, discount rates, and macro expectations
- The yield curve is a cross section of prices every month
- So we need notation for both the cross section and the dynamics
- Main question: when the curve moves, is it expected short rates or term premia?

- Part I: bond yields, forward rates, and holding returns
- Part II: the expectations hypothesis as a benchmark
- Part III: what Fama and Bliss ([1987](#)) and Campbell and Shiller ([1991](#)) found
- Part IV: Nelson-Siegel and why Diebold and Li ([2006](#)) made it useful for forecasting

Part I: Building Blocks

Zero-Coupon Bonds and Yields

- $P_t^{(n)}$ be the time- t price of an n -period zero-coupon bond ([Gürkaynak and Wright 2012](#))
- It pays \$1 at time $t + n$ (we are abstracting away from defaults)

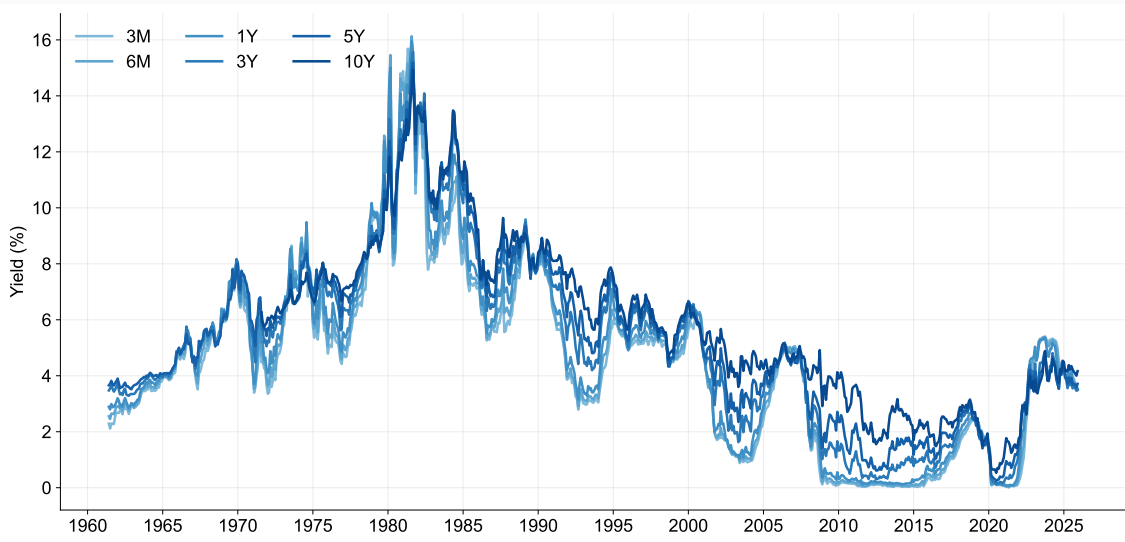
$$P_t^{(n)} = e^{-ny_t^{(n)}}, \quad y_t^{(n)} = -\frac{1}{n} \log P_t^{(n)}$$

- $y_t^{(n)}$ is the annualized continuously compounded yield
- Think about zero-coupon bonds are Arrow securities – they are our building block
- In most cases these yields are *not* observed directly. Why?

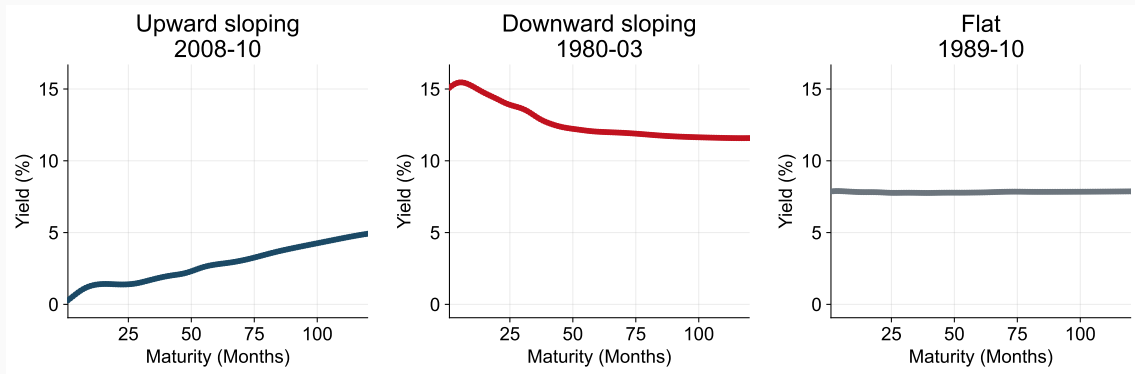
The Yield Curve

- The yield curve is the mapping $n \mapsto y_t^{(n)}$
- Same date, many maturities
- Long and short yields need not move one-for-one
- The curve can be upward sloping, downward sloping, flat, humped, ...
- There is a huge literature of how yields move over time and across n

US Treasury Yields Over Time



Yield Curve on Different Days



- The yield curve can look very different on different days
- Most frequently: upward sloping, and concave

Forward Rates

- Suppose you want to invest money for 2 years
- Option 1: buy a 2-year bond and get $y_t^{(2)}$ for the whole period
- Option 2: buy a 1-year bond, then reinvest the proceeds for another year

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- Return with option 2: $y_t^{(1)} + y_{t+1}^{(1)}$
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- Return with option 2: $y_t^{(1)} + y_{t+1}^{(1)}$
- The return for option 2 is *random*, of course
- The 1-year forward rate between $t + 1$ and $t + 2$ is the rate that makes the two options have the same expected return

$$f_t^{(1,1)} = 2y_t^{(2)} - y_t^{(1)}$$

Forward Rates:

- In general, the forward rate for an m -period investment starting n periods ahead is

$$f_t^{(n,m)} = \frac{1}{m} \left[(n+m)y_t^{(n+m)} - ny_t^{(n)} \right]$$

- When $n = 0$ and $m = 1$, this reduces to the short rate $y_t^{(1)}$
- Forward rates carry the same information as yields. Why?
- When $m \rightarrow 0$, we call it the instantaneous forward rate, $f_t^{(n)}$
- This is the “slope” of the yield curve at date t around maturity n

Yield = Average Forward Rates

For discrete one-year forwards,

$$y_t^{(n)} = \frac{1}{n} \sum_{i=1}^n f_t^{(i-1,1)}$$

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- A long yield is an average of future forward rates
- So the yield curve is already a summary of expected future financing conditions
- But it may also embed risk premia
- Notice that the equation above is a *tautology*

Holding-Period Returns

- Buy an n -period bond at t
- Sell it one period later as an $(n - 1)$ -period bond

$$r_{t+1}^{(n)} = \log P_{t+1}^{(n-1)} - \log P_t^{(n)} = ny_t^{(n)} - (n-1)y_{t+1}^{(n-1)}$$

- This is the realized one-period log return ([Gürkaynak and Wright 2012](#))
- Of course, we are abstracting away from taxes, transaction costs, and other frictions
- This return is random because $y_{t+1}^{(n-1)}$ is random

$$xr_{t+1}^{(n)} = r_{t+1}^{(n)} - y_t^{(1)} = ny_t^{(n)} - (n-1)y_{t+1}^{(n-1)} - y_t^{(1)}$$

- Compare the long bond with the risk-free one-period investment
- If $xr_{t+1}^{(n)} > 0$, the long bond beat the short rate over that period
- Empirical work often studies these excess returns directly
- Whenever people talk about “bond risk premia”, they usually mean expected excess returns

Part II: The Expectations Hypothesis

Definition (Expectations Hypothesis)

Long yields equal expected future short yields plus a maturity-specific constant:

$$y_t^{(n)} = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t(y_{t+i}^{(1)}) + c_n$$

- c_n can change with n but not with t
- Time variation in yields should mainly reflect changing expectations of future short rates
- We call $y_t^{(n)} - \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t(y_{t+i}^{(1)})$ the “term premium”
- The EH implies that this term premium is constant over time

EH can be written as (you should prove this at home)

$$\mathbb{E}_t \left[x r_{t+1}^{(n)} \right] = \mu_n$$

- Expected excess log return on an n -period bond is constant
- EH rules out time-varying expected bond premia
- Under EH, there is no reason for *any* time- t variable forecast returns

Questions?

Part III: Early Empirical Tests

Why These Regressions?

- EH is a conditional statement, so we test conditional forecasts
- Project future yield changes or excess returns on today's spreads
- Benchmark under EH: slope coefficients should be close to 1

$$y_{t+m}^{(m)} - y_t^{(m)} = a + b (y_t^{(n)} - y_t^{(m)}) + u_{t+m}, \quad xr_{t+1}^{(n)} = a + b [f_t^{(n-1,1)} - y_t^{(1)}] + u_{t+1}$$

- If b is far from 1, the EH benchmark is in trouble

Fama and Bliss (1987) focus on the forward-spot spread:

$$f_t^{(n-1,1)} - y_t^{(1)} = \mathbb{E}_t \left[y_{t+n-1}^{(1)} - y_t^{(1)} \right] + \mathbb{E}_t \left[x r_{t+1}^{(n)} \right]$$

Fama-Bliss: Forward-Spot Decomposition

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- It can be decomposed into expected future short-rate changes
- And expected excess returns on longer bonds
- Their regressions ask: which margin actually moves?

Fama and Bliss 1987: Main Result

$$xr_{t+1}^{(n)} = a + b \left[f_t^{(n-1,1)} - y_t^{(1)} \right] + u_{t+1}$$

Dependent Variable	a	$s(a)$	b	$s(b)$	R^2
$xr_{t+1}^{(2)}$	-0.21	0.41	0.91	0.28	0.14
$xr_{t+1}^{(3)}$	-0.51	0.68	1.13	0.37	0.11
$xr_{t+1}^{(4)}$	-0.91	0.92	1.42	0.45	0.11
$xr_{t+1}^{(5)}$	-1.06	1.31	0.93	0.53	0.05

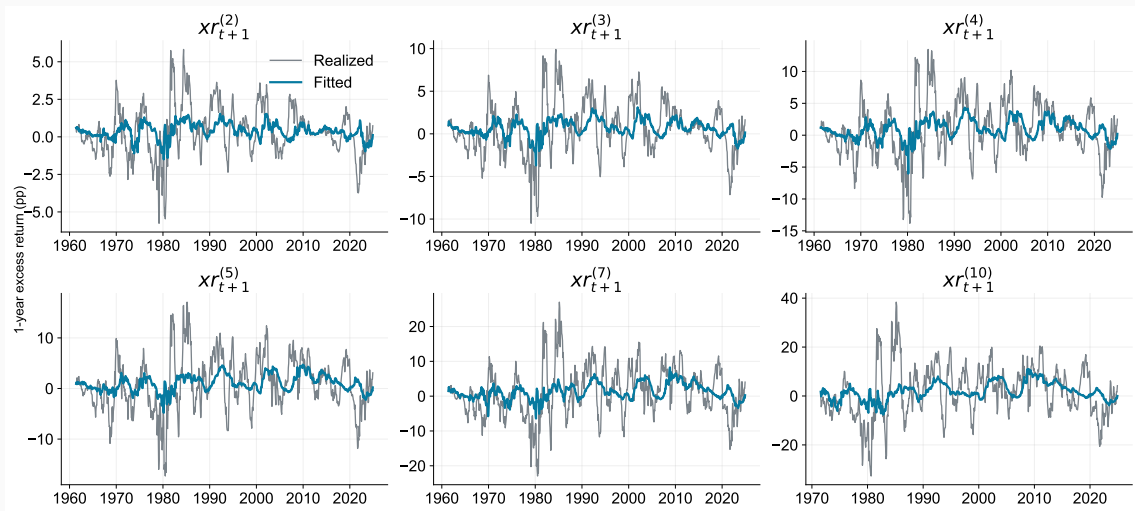
- They found that forward rates forecast excess returns
- Forward rates are known at time t ... so it's bad news for EH
- Data is from 1964 to 1984

Fama-Bliss: Updated Regression

Dependent Variable	a	$s(a)$	b	$s(b)$	R^2
$xr_{t+1}^{(2)}$	0.08	0.18	0.76	0.22	0.09
$xr_{t+1}^{(3)}$	0.02	0.35	0.95	0.27	0.10
$xr_{t+1}^{(4)}$	-0.19	0.51	1.19	0.31	0.12
$xr_{t+1}^{(5)}$	-0.18	0.67	1.16	0.35	0.09
$xr_{t+1}^{(7)}$	-0.55	0.88	1.40	0.34	0.10
$xr_{t+1}^{(10)}$	-0.79	1.48	1.69	0.44	0.11

- We redo the Fama-Bliss regression on newer monthly zero-coupon data
- The signal is still the forward-spot spread, $f_t^{(n-1,1)} - y_t^{(1)}$
- Same results 40 years later (1962-2024 sample)!

Realized vs Fitted Excess Returns



Questions?

Campbell and Shiller 1991: Spread Restrictions

Campbell and Shiller (1991) study yield spreads directly

$$s_t^{(n)} \equiv y_t^{(n)} - y_t^{(1)}$$

- Expected excess returns must satisfy, by definition:

$$xr_{t+1}^{(n)} = n s_t^{(n)} - (n-1) [y_{t+1}^{(n-1)} - y_t^{(1)}]$$

- Under EH, it should hold that

$$s_t^{(n)} = \mu_n + (n-1) \mathbb{E}_t [y_{t+1}^{(n-1)} - y_t^{(n)}]$$

Some Possible Tests

- Suppose you run the following regression (call it *Long Rate Regression*):

$$(n-1) \left(y_{t+1}^{(n-1)} - y_t^{(n)} \right) = \alpha + \beta s_t^{(n)} + u_{t+1}$$

- Under EH, the slope β should be close to 1!

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- Another restriction is given if you apply the definition of $s_t^{(n)}$ to the EH equation:

$$s_t^{(n)} = \theta_n + \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t \left[y_{t+i}^{(1)} - y_t^{(1)} \right] = \theta_n + \sum_{i=1}^{n-1} \left(1 - \frac{i}{n} \right) \mathbb{E}_t \Delta y_{t+i}^{(1)}$$

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- Now you run the following (call it *Short Rate Regression*):

$$\sum_{i=1}^{n-1} \left(1 - \frac{i}{n} \right) \Delta y_{t+i}^{(1)} = \alpha + \beta s_t^{(n)} + u_{t+n-1}$$

- Again, EH says β should be close to 1

Campbell and Shiller 1991: Main Result

Variable	Long bond maturity (n), quarterly intervals					
	2	3	4	8	20	40
Long-yield changes	-1.277 (0.319)	-1.082 (0.432)	-1.170 (0.489)	-1.505 (0.619)	-2.283 (0.736)	-3.133 (0.895)
Short-rate changes	0.252 (0.137)	0.232 (0.143)	0.253 (0.132)	0.427 (0.142)	0.835 (0.117)	0.554 (0.126)

- This is using quarterly data from 1961 to 2013
- This is not from the original paper, but from Campbell's book (Chap 8)
- Bottom-line: all coefficients are *very* far from 1

Campbell and Shiller 1991: Updated LW Table

Variable	Long bond maturity (n), monthly intervals				
	6m	1y	2y	5y	10y
Long-yield changes	-0.499 (0.587)	-0.762 (0.628)	-1.053 (0.694)	-1.910 (0.840)	-3.150 (1.005)
Short-rate changes	0.407 (0.183)	0.470 (0.185)	0.576 (0.214)	0.732 (0.224)	0.778 (0.132)

- Monthly LW zero-coupon yields; 6m-5y columns start in 1961:06, the 10y column in 1971:08
- Same message as Campbell-Shiller 35 years later
- Any intuition regarding the sign of these effects?

One Message from Both Papers

- The EH is a bad model for yield-curve dynamics
- It generates restrictions that are violated in the data (then, and now)
- The natural implication here is that there are *time-varying term premia*
- But now the problem is harder...
- We need theories for *both* expected short rates and term premia

Part IV: Nelson-Siegel and Diebold-Li

- The EH is not a reasonable model for yield-curve dynamics
- We do not just want one regression per maturity
- We want a parsimonious model for the whole cross-section of yields
- And then a simple model for its dynamics
- Sometimes we need to forecast things, not just explain them in-sample...
- This is the problem Diebold and Li ([2006](#)) tackles directly

Nelson-Siegel Forward Curve

Start from a smooth forward-rate curve ([Nelson and Siegel 1987](#); [Diebold and Li 2006](#)):

$$f_t^{(\tau)} = \beta_{1t} + \beta_{2t}e^{-\lambda\tau} + \beta_{3t}\lambda\tau e^{-\lambda\tau}$$

- Three components, one decay parameter λ
- This is the solution to a 2nd-order constant-coefficient ODE for $f(\tau)$
- Simplest smooth specification that can generate level, monotone, and humped shapes
- Only 4 parameters, but nests flat ($\beta_2 = \beta_3 = 0$), exponential ($\beta_3 = 0$), and humped curves

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- How would you construct y_t^τ from this forward curve?

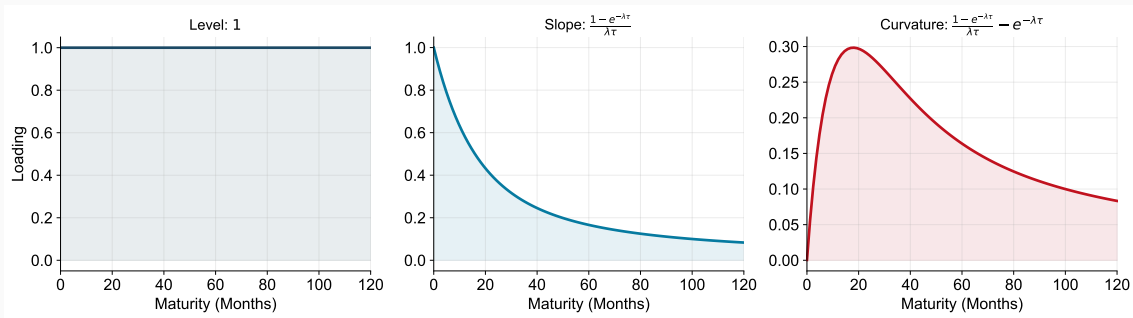
The Implied Yield Curve

Averaging forward rates gives

$$y_t^{(\tau)} = \beta_{1t} + \beta_{2t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_{3t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)$$

- Same time-varying factors at every maturity
- Different maturities load differently on them
- Proper factor model for yields

How This Basis Looks Like



- Loading on β_{1t} is 1 at all maturities
- Loading on β_{2t} starts at 1 and decays to 0
- Loading on β_{3t} starts at 0, rises, then decays

What Each Factor Does

- β_{1t} shifts the whole curve, but also notice that $\lim_{\tau \rightarrow \infty} y_t^{(\tau)} = \beta_{1t}$ for any $\lambda > 0$
- β_{2t} moves the short end more than the long end
- β_{3t} bends the middle of the curve
- A small number of factors captures the main co-movements in yields

- Fix λ to an empirically motivated value
- Diebold and Li (2006) use $\lambda = 0.0609$
- Each month, estimate β_{1t} , β_{2t} , β_{3t} from the cross section of yields
- Then treat those estimated coefficients as factor time series
- Much easier to interpret than an unrestricted factor model
- No rotation indeterminacy

Quality of Fit

6-Month



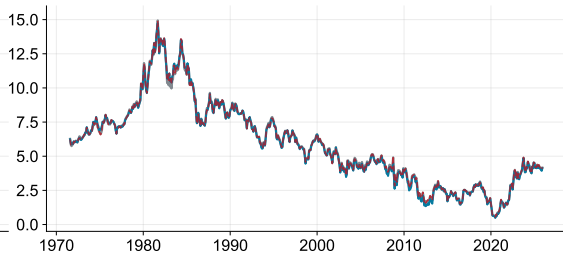
1-Year



5-Year



10-Year



Forecast the Factors, Forecast the Curve

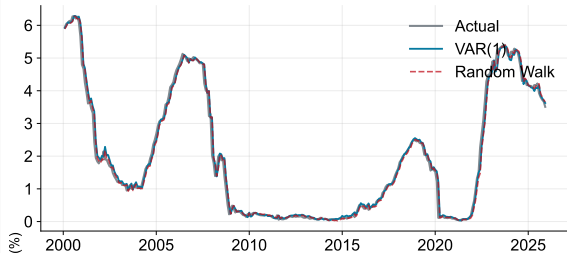
Let the factors follow simple dynamics:

$$\beta_t = c + A\beta_{t-1} + u_t$$

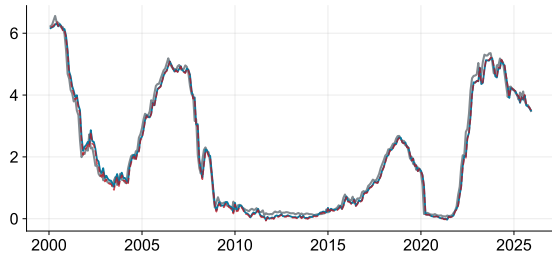
- Forecast the factors with AR or VAR models
- Plug factor forecasts into the Nelson-Siegel loadings
- Result: a forecasting model for the entire curve ([Diebold and Li 2006](#))
- Key move: turn a cross-sectional decomposition into a dynamic forecasting system
- Common practice in the literature: FAVAR models with NS factors

Empirical Results

6-Month



1-Year



5-Year

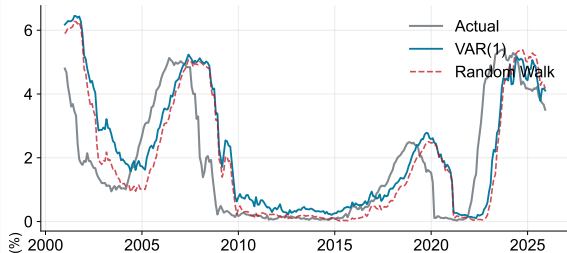


10-Year

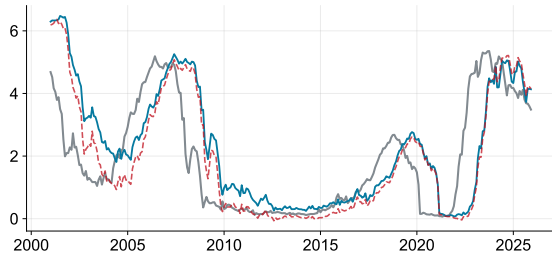


Empirical Results: 12-Month Horizon

6-Month



1-Year



5-Year



10-Year



Out-of-Sample MSE

Maturity	1-Month Horizon			12-Month Horizon			
	VAR(1)	RW	Ratio	VAR(1)	VAR(6)	RW	Ratio ₁
6-Month	0.0394	0.0413	0.954	2.1286	2.5517	2.1146	1.007
1-Year	0.0567	0.0562	1.009	2.1546	2.5196	1.9496	1.105
5-Year	0.0910	0.0794	1.146	1.3357	1.5756	0.9587	1.393
10-Year	0.0786	0.0819	0.960	0.7581	0.8893	0.6429	1.179

- It's super tough to beat a RW
- In the paper they could do it for shorter horizons sometimes
- But it's always very hard!
- HUGE literature about macro forecasting!

Questions?

- Part I gave us the language: yields, forwards, and holding returns
- Part II gave us the benchmark: EH says spreads should reflect expected short rates, not time-varying premia
- Part III showed the classic failure: forwards and spreads forecast returns too well
- Part IV gave a practical alternative: summarize the whole curve with level, slope, and curvature, then forecast those factors
- Next class: the dance between expected rates and term premia

Readings and References i

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